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Subject: Engineering Physics - IA 1 | Section: B

Q1

1st law - a body at rest remains at rest unless a force acts on it.

2nd law - F = ma, force gives acceleration to the body.

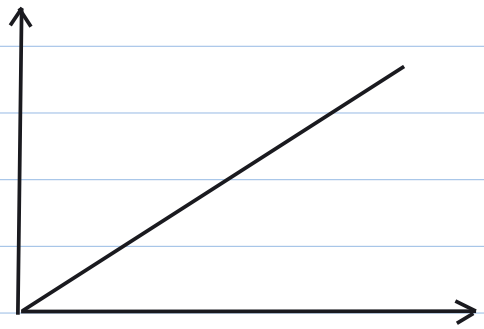
Q2

a = v/t = 20/5 = 4

s = 1/2 at^2 = 1/2 x 4 x 5 = 10 m

Q3

Ohms law: V = IR. Voltage is proportional to current.



Q4

KE = 1/2 mv^2 is energy of motion. PE = mgh.

mgh = 1/2 mv^2

v = 2gh = 2 x 10 x 20 = 400 m/s